MTH1003

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

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Mathematical Modelling

Module Convener: Dr Prof. John Thuburn

You have 24 hours to complete this paper from the time of its release
Intended duration 2 hours

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is an **OPEN BOOK** examination

SECTION A

1. (a) A particle of mass $m = 1$ is acted upon by a force

$$\mathbf{F} = \frac{1}{2}\mathbf{i}(t+1)^{-3/2} - 2\mathbf{j}\cos t.$$

If the particle is initially at $\mathbf{r} = 2\mathbf{j} - 2\mathbf{i}$ with velocity $-\mathbf{i}$, find its subsequent velocity and position. At what times $t > 0$ is its position vector perpendicular to its velocity vector? (10)

(b) For a particle moving in one dimension along the x-axis, show that the acceleration $\ddot{x}$ can be expressed as the x-derivative of a certain quantity. Hence derive the energy equation for the simple harmonic oscillator, whose equation of motion is

$$\ddot{x} + \omega^2 x = 0,$$

where ω is a constant. (6)

(c) State Hooke's law for the tension in an elastic string, defining any mathematical variables that you introduce.

A particle of mass m is attached to one end of a light elastic string and the other end is fixed to a rigid support. The string hangs vertically under gravity, where g is the gravitational acceleration. Obtain the equation of motion for vertical displacements of the particle in terms of its distance z from the support. Find the equilibrium position of the particle.

When in its equilibrium position the particle is given an upward velocity U; under what condition on U does the string become slack during the subsequent motion? (14)

(d) Suppose we wish to find a numerical solution to a nonlinear equation of the form $x = g(x)$. Write down the fixed point iteration formula for the solution of this problem. State a necessary condition for the convergence of the fixed point iteration to the true solution.

Write a short Matlab code that will execute 8 iterations of the fixed point iteration scheme for the case $g(x) = \cos(x/3)$, starting from the first guess $x^{(0)} = 1$, and display the result. Be careful to use correct syntax. (10)

(e) Sketch the nullclines of the dynamical system

$$\dot{x} = x(x^2 - 4)\mathrm{e}^{-y}; \qquad \dot{y} = y(1 - x - y)\mathrm{e}^{-x}.$$

Hence find all the equilibrium points for real x and y. (10)

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SECTION B

2. A particle is launched from the origin on flat ground in the x-z plane with initial speed V at an angle α to the horizontal. Let g be the gravitational acceleration.

   (a) Assuming the particle meets no obstacle, write down its x and z coordinates as functions of time. Hence obtain the trajectory equation of the particle. (5)

   (b) Now suppose that there is a wall of height H at position $x = 2H$ and that the launch speed of the partice is $V = 2\sqrt{gH}$. Determine the range of angles α for which the particle goes over the wall. Find the set of values of $x > 0$ at which the particle can hit the ground i.e. the point $(x, 0)$; consider points on both sides of the wall. (12)

   (c) A second wall of height H is now positioned at $x = X < 2H$ such that the particle cannot pass over both walls for any angle α. What is the largest possible value of X? (8)

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3. (a) A particle with position $x(t)$ undergoes damped harmonic motion governed by

$$\ddot{x} + 2k\dot{x} + \omega^2 x = 0,$$

   with initial conditions $x = 2$, $\dot{x} = 2$. For the two cases

$$\begin{array}{lll} \text{(i)} & k = 3, & \omega^2 = 25, \\ \text{(ii)} & k = 3, & \omega^2 = 9, \end{array}$$

   obtain the solution $x(t)$ and state what kind of damping occurs. (14)

   (b) Consider the damped oscillatory motion described by the equation

$$\dot{x} = (a + \mathrm{i}b)x,$$

   where a and b are real with $a < 0$, and with initial condition $x(0) = x_0$. Write down the exact solution for $x(t)$.
   Write down the forward Euler time stepping scheme with time step h for the solution of the above equation. Calculate the amplification factor for the forward Euler scheme. Hence show that the forward Euler scheme is stable provided

$$-1 - \{1 - (bh)^2\}^{1/2} \leq ah \leq -1 + \{1 - (bh)^2\}^{1/2}.$$

   (11)

   [**25**]

4. (a) Find all the equilibrium points of the following dynamical system, and classify their stability type:

$$\dot{x} = (x-1)(3-y), \qquad \dot{y} = (y-1)(x-2).$$

(8)

(b) Giving appropriate justification, sketch the trajectories near each equilibrium point, and hence in the phase plane. (8)

(c) Find a conserved quantity for this dynamical system. Explain briefly why the system can never evolve from $x > 1$ to $x < 1$. (9)

[**25**]